

MUDITHA PRIYASAD

Mechanical Design, Automation & R&D Engineer

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PROFESSIONAL SUMMARY

Hands-on Mechanical Design, Automation and R&D Engineer with experience in machine design, factory automation, PLC-based controls, pneumatic systems, troubleshooting and rapid prototyping. Develops production machines from concept to commissioning, connecting mechanical design with automation and manufacturing needs.

Highlights: 15+ practical projects | 4 industrial environments | PLC, pneumatic, CAD and prototyping skills

CORE SKILLS

Machine Design: Jigs, fixtures, frames, mechanisms, SolidWorks/3D CAD, 2D fabrication drawings, DfMA/DfAM

Automation & Controls: Siemens S7-200, Delta, Xinje and Mitsubishi PLCs; sensors, actuators, pneumatics, servo mechanisms, wiring and control panels; machine commissioning

Production Improvement: Troubleshooting, root-cause analysis, mechanical evaluation, preventive & predictive maintenance, process optimization, cycle-time reduction, TPM, 5S, Kaizen

Prototyping & R&D: 3D printing, CNC laser systems, G-code, workshop fabrication, design validation, fit/function testing and low-cost prototyping

Digital Tools: Raspberry Pi, Python, C++, Microsoft Office, dashboards and AI-assisted engineering workflows

Software: SOLIDWORKS, AutoCAD, STEP 7 Micro/WIN, Delta ISPSoft, Mitsubishi GX Works, Xinje PLC Programming Software, Arduino IDE

PROFESSIONAL EXPERIENCE

Consultant Engineer - Mechanical Design & Automation

MARQO Industries Pvt Ltd | 2024 - Present

- Increased fan base cement-filling throughput 6x, from 1 to 6 units/min, by developing and commissioning a semi-automatic cement filling machine for faster and more consistent filling than manual work.
- Increased B22 eyelet punching speed 3x and reduced operator effort by designing an ergonomic automated punching mechanism.
- Improved fan motor assembly efficiency and reduced operator strain by creating a guided fixture with hands-free oil application and assisted fastener tightening.
- Managed setup, commissioning and maintenance troubleshooting of plug-point packing and spot-welding equipment, coordinating project tasks and producing prototype/support parts for production improvement.
- Established the engineering and automation function from scratch by creating workshop guides, department software, standardized tools and core processes.

Founder / Designer

MTone Designers | 2021 - Present

- 3D printing, CNC laser engraving and machine and product design.
- Build and test custom 3D printer and CNC laser machines.

Engineering Intern

MAS Active Shadowline | 2024 (9 months)

- Contributed to TPM, 5S and Kaizen work by preparing machine standards, checklists, SOPs, dashboards and handover documents.
- Sustained monthly electricity, water, waste and KPI records while temporarily covering Sustainability Executive duties.
- Prepared CAD layouts, tool/equipment lists, budget inputs, training material, visual labels and production-support documents.

Automation Engineering Intern

Effective Engineering & Automation Pvt Ltd | 2023 (6 months)

- Built practical capability in industrial automation, troubleshooting, wiring, PLC programming, equipment testing and documentation.
- Worked on machine and test-equipment projects including a 3D printer, line-follower robot, screen printer, concrete mixers and glove-making robot support tasks.

Autonomation Engineering Intern

MAS Kreedaa Vaanavil | 2019 (3 months)

- Applied pneumatic circuit design, CAD, PLC ladder logic, Arduino programming, power tools and lathe-machine operation.
- Supported boiler monitoring, bonding-tape machine development and fabric-picking mechanism research.

KEY PROJECTS

- **Fan Base Cement-Filling Machine** – semi-automatic filling machine (design through commissioning).
- **B22 Eyelet Puncher** – ergonomic automated punching mechanism for faster, lower-effort operation.
- **Fan Motor Assembly Fixture** – guided fixture with hands-free oiling and assisted fastening.
- **Plug-Point Packing & Spot-Welding** – commissioning, troubleshooting and support tooling.
- **PCB Drilling Machine** – CNC drilling rig for NDM Electrical (18 PCBs/setup, improved accuracy & repeatability).
- **Bulb Assembly Machines** – setup and troubleshooting of two assembly machines for NDM Electrical.
- **Custom 3D Printer** – frame, motion control and electronics integration with calibration and testing.
- **1W LED Bulb Assembly Machine** – automated assembly concept to reduce handling and improve consistency.

EDUCATION

BSc (Hons) in Mechanical Engineering

University of Jaffna - Faculty of Engineering | October 2017 - February 2024

CERTIFICATIONS

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| • CSWA – Mechanical Design | • CSWA-AM – Additive Manufacturing |
| • Maintenance of Industrial Plant & Machinery – NERDC | • PLC Training – Univ. of Moratuwa, 2025 |
| • Industrial Motor Operation & Control – NERDC | • Advanced Injection Molding – IDB |

ACHIEVEMENTS & ACTIVITIES

- Bronze Award - Best Display of Engineering Projects, Techno Sri Lanka 2023
- Participant - AIR TATTOO Exhibition 2024
- Participant - SLT Robot Battles 2019
- Organizer - UOJ BATTLEGROUNDS 2020
- Languages - English, Sinhala



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